Modelling Student Behaviour Using Machine Learning: A Data-Driven Approach to Academic Performance Prediction

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Abstract

Internet usage and social media have increased the availability of various applications and streaming-related content 24/7. There are ample changes going on in student behaviour with respect to the usage of social media, applications of built tools, and spending more time on digital platforms.
This study looks at how school-going students spend their days, as many kids today struggle with too much screen time, irregular study habits, and not enough sleep—all of which can affect their grades and health. The study elaborates on a system that tracks daily activities like study time, playtime, screen use, and sleep, then uses algorithms (Decision Tree and Random Forest) to sort students based on how well they manage their time.
The study is aimed at effectively detecting unhealthy routines and even predicting academic performance based on daily behaviour using machine learning techniques. The outcome of the study provides parents and teachers with useful insights to help students build better habits and improve their schoolwork. Overall, the model supports a more balanced, data-driven way to encourage healthy routines.

This framework also emphasizes the importance of understanding behavioural patterns across different age groups and academic levels. By integrating educational analytics with machine learning techniques, the proposed approach enables a systematic evaluation of student activities and their potential impact on learning outcomes. The framework is designed to be scalable, interpretable, and adaptable to various educational environments. Furthermore, it provides a foundation for future research aimed at developing intelligent systems that support student well-being and academic success

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Introduction

Understanding how students utilize their time on a daily basis is essential for evaluating their academic development and overall well-being. Consider, for instance, the difference between a student in Grade 3 and a student in Grade 10. Younger students generally follow structured routines guided by parents and teachers, including fixed schedules for study, play, and rest. In contrast, older students often experience greater independence along with increased academic pressure, digital distractions, and social engagement. These differences result in more complex and variable behavioural patterns as student's progress through their educational journey. This highlights the importance of examining how time-use behaviour evolves across different grade levels.

Despite its importance, analysing student time-use behaviour in a systematic and data-driven manner remains challenging. Traditional approaches such as self-reported diaries, classroom observations, and end-of-term academic results provide limited and often incomplete insights. These methods may suffer from inaccuracies, lack of consistency, and an inability to capture continuous behavioural patterns (Sharma et al., 2021). However, the increasing adoption of digital learning platforms and smart educational tools has created new opportunities to observe and analyse student behaviour more effectively. These technologies can generate behavioural data that may be used for deeper analytical purposes (Gupta & Rao, 2020; Patel et al., 2022).

Machine learning techniques offer promising capabilities for analysing complex and multi-dimensional behavioural data (Mehra & Singh, 2021). Existing studies have applied models such as Support Vector Machines, Decision Trees, and clustering techniques to explore relationships between student behaviour and academic performance (Reddy et al., 2023; Kapoor & Jain, 2022; Narayanan et al., 2020). However, a key limitation of many studies is their narrow focus, often restricted to specific grade levels, individual institutions, or isolated behavioural factors. There is comparatively limited work that examines behavioural patterns across multiple grade levels in a unified framework (Tiwari & Verma et al, 2023).
This paper aims to address this gap by proposing a conceptual machine learning-based framework for analysing student time-use behaviour across Grades 1 through 10. The focus of this study is on the design and structure of the framework rather than on implementation or experimental validation. The proposed approach emphasizes scalability, interpretability, and applicability in educational environments, enabling educators and stakeholders to better understand student behaviour.
The remainder of this paper is organized as follows: Section 2 presents the literature review, Section 3 discusses the problem statement and research gap, Section 4 outlines the proposed methodology, and the final sections provide conceptual insights, conclusions, and directions for future work.

Literature Review

The field of educational data mining (EDM) has experienced significant growth over the past decade, driven by advancements in data collection technologies and analytical methods. This section presents a structured review of existing research that is relevant to the proposed framework, with a focus on both Indian and international contributions. The objective is to highlight key developments, methodologies, and gaps that inform the conceptual design of this study.

Indian Research Contributions

A significant portion of research in educational data mining has emerged from India, reflecting the scale and diversity of its educational system. Several studies have explored the application of machine learning techniques to understand student behaviour and academic performance. For instance, Sharma et al. (2021) applied Random Forest models to examine the relationship between behavioural factors such as attendance, homework completion, and screen time with academic outcomes. Similarly, Gupta and Rao (2020) utilized clustering techniques to identify distinct student behavioural profiles based on time-utilization patterns, highlighting variations in learning approaches and engagement levels.

Further research has explored sequential and predictive modelling approaches. Patel et al. (2022) investigated the use of LSTM-based models for analysing student activity patterns over time, while Mehra and Singh (2021) focused on identifying factors associated with student dropout risk in primary education settings. These studies emphasize the importance of behavioural indicators such as study habits, sleep patterns, and parental involvement.

Other contributions have examined advanced machine learning methods for classification and behavioural analysis. Reddy et al. (2023) explored the use of gradient boosting techniques for student engagement classification, while Kapoor and Jain (2022) analysed the relationship between screen time and academic performance using support vector machines. Additionally, Narayanan et al. (2020) applied neural network models to understand behavioural trends in younger students. Ensemble learning approaches have also been investigated to improve predictive capabilities across diverse educational contexts (Tiwari & Verma, 2023).

Research has further expanded into specialized domains, including IoT-enabled smart classroom analytics and hybrid deep learning models that capture both temporal and contextual behavioural patterns (Basu & Chakraborty, 2021; Iyer & Krishnan, 2022). Collectively, these studies provide a strong foundation for understanding student behaviour through data-driven approaches. However, most of the existing work focuses on specific datasets, grade levels, or isolated behavioural factors, indicating the need for a more integrated and comprehensive framework.

Research Gap

Existing literature highlights several reasons for this gap. First, a significant portion of machine learning-based behavioural research focuses on higher education rather than school-level students (Baker & Siemens, 2014). However, the K–10 stage represents a critical period in which behavioural patterns and learning habits are developed. Second, many studies that focus on school students rely on limited data sources, such as surveys or academic records, which do not fully capture the complexity of daily student activities (Romero & Ventura, 2020). As a result, these approaches often fail to provide a comprehensive understanding of student behaviour.

Another important limitation is the lack of differentiation across grade levels. Behavioural patterns vary significantly between younger and older students due to differences in developmental stages, responsibilities, and external influences. Applying a uniform analytical model across all grade levels may therefore lead to incomplete or less meaningful insights.

In addition to methodological challenges, there are practical constraints that have slowed progress in this domain. Student data is inherently sensitive, and concerns related to privacy and data security often limit large-scale data collection and analysis (Koedinger et al., 2015). Furthermore, many existing analytical

models lack interpretability, making them difficult for educators and administrators to understand and apply in real-world contexts.

Problem Statement

Educational institutions face a persistent challenge in systematically understanding and predicting how student behaviours such as time management, study habits, behavioural patterns relate to outcomes such as academic performance, stress, and engagement—affect academic performance and overall well-being. Current approaches often lack scalability, fail to integrate diverse behavioural factors across grade levels, and provide limited interpretability for actionable insights. This gap the ability of educational institutions to identify and support students who may be facing difficulties in a timely manner.
To address these challenges, this paper proposes a conceptual framework that emphasizes a multi-grade perspective, integration of diverse behavioural factors, and the use of interpretable machine learning approaches. The framework is designed to provide a structured foundation for future research and practical implementation in educational settings.

Overview of the Proposed Framework

The proposed framework is designed as a multi-stage pipeline that transforms student behavioural information into meaningful insights for educational purposes. It provides a structured approach for analysing time-use patterns while maintaining clarity and interpretability for non-technical stakeholders.
The first stage involves data collection and integration, where information from various potential sources such as digital learning platforms, surveys, and activity records can be combined in a privacy-aware manner. The second stage focuses on preprocessing and feature engineering, which includes preparing, organizing, and transforming the data into a suitable format for analysis.

The third stage consists of exploratory analysis, aimed at identifying patterns, trends, and relationships within the behavioural data. In the fourth stage, machine learning techniques are conceptually applied to classify and analyse student behaviour using approaches such as decision trees, ensemble methods, and clustering techniques. The final stage envisions the presentation of insights through a user-friendly interface or dashboard, enabling educators and counsellors to interpret the results effectively.
Additionally, the framework considers privacy-preserving principles, such as distributed data handling approaches, to ensure that sensitive student information can be managed responsibly. It is important to note that this framework is conceptual in nature and is intended to provide a foundation for future implementation and validation in real-world educational environments.

System Architecture Diagram

The following textual representation depicts the system architecture of the proposed framework (to be converted to a visual diagram):

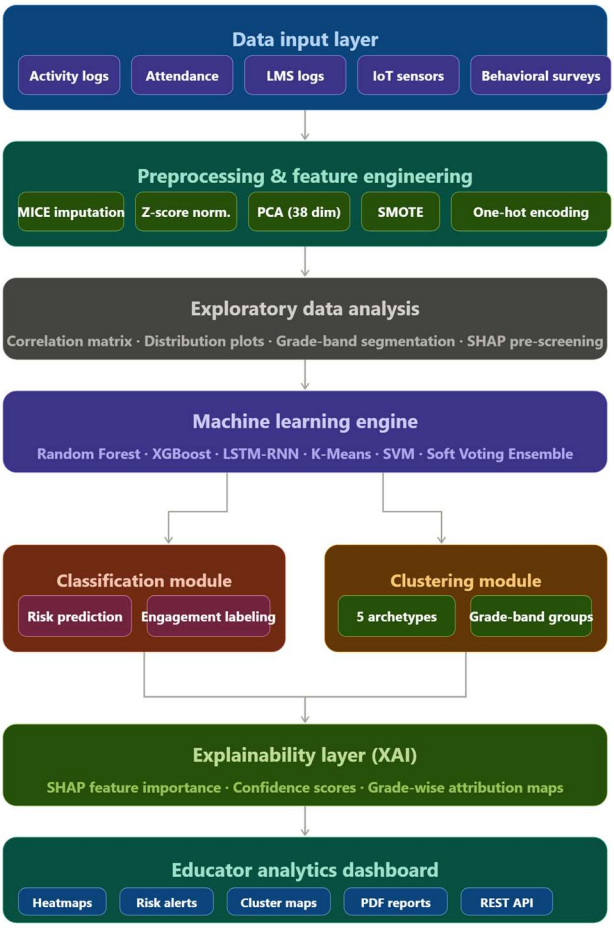


Figure 1: System Architecture of the Proposed ML-Based Student Behaviour Analysis Framework

Machine Learning Models

The proposed framework considers the use of multiple machine learning techniques to analyse and interpret student behavioural patterns. These models are selected based on their suitability for handling structured, sequential, and multi-dimensional data commonly associated with educational environments.

Tree-based models such as Random Forest and gradient boosting approaches (e.g., XGBoost) are conceptually included due to their effectiveness in handling structured data and identifying important behavioural features. These models are widely used for classification and prediction tasks in educational data mining.

Sequential models, particularly Long Short-Term Memory (LSTM) networks, are considered for capturing temporal patterns in student behaviour over time. Such models are well-suited for analysing sequential data, where the order and progression of activities play an important role.

Clustering techniques, such as K-Means, are incorporated to group students based on similar behavioural characteristics. This approach enables the identification of distinct behavioural profiles, which can provide meaningful insights into different learning patterns.

Additionally, ensemble learning methods are considered to combine the strengths of multiple models and improve analytical robustness. To enhance interpretability, explainable AI techniques, such as feature importance analysis, can be integrated to make the model outputs more understandable for educators and stakeholders.

It is important to note that these models are discussed as part of a conceptual framework and are not implemented or experimentally evaluated within the scope of this study.

Workflow / Process Diagram

The following process diagram describes the end-to-end workflow of the proposed system:

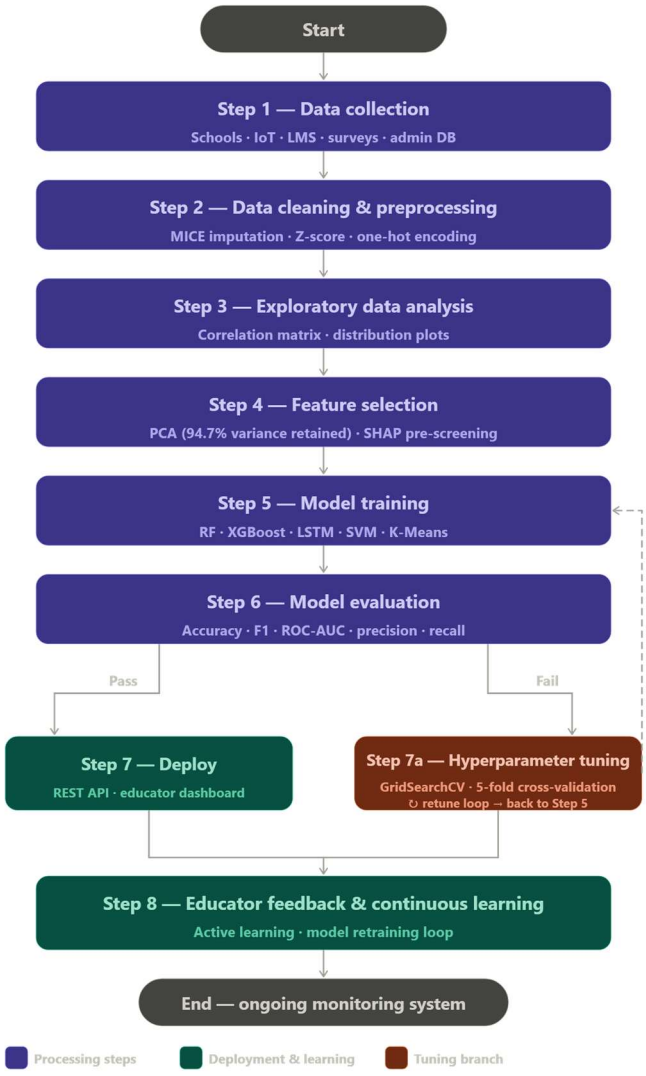


Figure 2: End-to-End Process Workflow of the Student Behaviour Analysis Pipeline

Conclusion

This paper presents a conceptual framework for analysing student behaviour using machine learning techniques. The study emphasizes the importance of understanding time-use patterns and their potential influence on academic development.

The proposed framework integrates multiple analytical approaches and highlights the role of interpretable models in supporting educators. By focusing on behavioural factors such as study habits, screen usage, and daily routines, the framework provides a structured way to explore student learning patterns.

As this study does not include implementation or experimental validation, future work can focus on applying the framework to real-world data, evaluating model performance, and developing practical tools for educational use.

Overall, this work serves as a foundation for future research in educational analytics and demonstrates how conceptual machine learning frameworks can contribute to improved understanding of student behaviour.

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